Definition 7.1

Let G be a group. A subset $H \subseteq G$ is a *subgroup* of G if it is a group under the operation in G.

Note. If G is a group, then the largest subgroup of G is the group G itself. The smallest subgroup of G is the group $\{e\}$ consisting of the identity element of G only.

Theorem 7.2

Let G be a group. A subset $H \subseteq G$ is a subgroup of G if and only if the following conditions are satisfied:

1) The identity element e belongs to H.
2) If $a, b \in H$ then $a \cdot b \in H$.
3) If $a \in H$ then $a^{-1} \in H$.

Exercise. The dihedral group D_4 has the following multiplication table:

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

Find all subgroups of D_4.

Definition 7.3

The *center* of a group G is a set $Z(G) \subset G$ consisting of elements that commute with all elements of G:

$$Z(G) = \{g \in G \mid ag = ga \text{ for all } a \in G\}$$

Exercise. Find the center of the dihedral group D_4.

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

Theorem 7.4

If G is a group then the center $Z(G)$ of G is a subgroup of G.

Definition 7.5

If G is a group and S is a non-empty subset of G, then $\langle S\rangle$ denotes the smallest subgroup of G containing all elements of S:

$$\langle S\rangle = \{a_1^{k_1} \cdot a_2^{k_2} \cdot \dots \cdot a_n^{k_n} \mid n \geq 1 \text{ and for each } i \text{ we have } a_i \in S \text{ and } k_i \in \mathbb{Z}\}$$

We say that $\langle S\rangle$ is the *subgroup of G generated by the set S*.

Note. If $a \in G$ then $\langle a\rangle = \{a^k \mid k \in \mathbb{Z}\}$.

Exercise. Find the subgroup $\langle 2\rangle$ in $\mathbb{Z}_{10}$.

Exercise. Find the subgroup $\langle 2\rangle$ in $\mathbb{Z}_9$.

Exercise. Find the subgroup $\langle V, R_{180}\rangle$ in D_4.

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

Recall:

- The order of a group G is the number of elements of G. It is denoted by $|G|$.

- The order of an element a of a group G is the smallest integer $n > 0$ such that $a^n = e$. It is denoted by $|a|$.

Theorem 7.6

Let G be a group, let $a \in G$ and let $\langle a \rangle$ be the subgroup of G generated by a. Then

$$|a| = |\langle a \rangle|$$